

Quantum Orthogonal Vision Transformers for Dark Matter Classification in Gravitational Lensing: A Data-Efficient Approach via Orthogonal Attention

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Abstract

We present **Quantum Orthogonal Vision Transformers (QOVT)**, a hybrid quantum-classical architecture that replaces the query/key/value attention projections in a Vision Transformer with orthogonal matrices parameterized by quantum circuits, following Cherrat et al. (2022). Applied to the DeepLense gravitational lensing benchmark, we evaluate two circuit variants: a Givens butterfly decomposition (1,536 trainable angles) and an RY+CNOT ring circuit (384 trainable angles, 6 qubits, $D = 2^6 = 64$). The RY+CNOT variant achieves $\text{AUC} = 0.9144$ at $N = 500$ samples/class— compared to 0.5134 for the Givens butterfly at the same data size— a gap consistent with the Caro et al. (2022) generalization bound: fewer trainable quantum parameters predict stronger generalization when training data is scarce. At full data, the Givens variant achieves $\text{AUC} = 0.9940$ on Model II, competitive with classical multi-head attention (0.9921). Crucially, the RY+CNOT circuit is directly deployable on NISQ hardware (IBM Quantum and equivalents) with no PennyLane overhead, running $10\times$ faster. These results establish orthogonal attention constraints as an effective geometric prior for low-data gravitational lensing classification.

1 Introduction

Strong gravitational lensing offers a unique window into dark matter substructure. When a background light source is lensed by a foreground mass distribution, the resulting arcs and Einstein rings carry morphological imprints of the dark matter model: axion dark matter, cold dark matter (CDM) subhalos, and smooth (no-substructure) halos each produce statistically distinguishable distortions [17]. Automated classification of these three scenarios from 64×64 pixel images is a core problem in the ML4SCI / DeepLense programme.

Recent deep learning methods have made substantial progress on the DeepLense benchmark. The state-of-the-art baseline is an MAE-pretrained Vision Transformer (MAE-ViT) [25] achieving $\text{AUC} \approx 0.97$ at full data, but this approach requires 2.72×10^6 parameters and a self-supervised pretraining phase. A central open question is whether *data-efficient* alternatives exist that do not depend on large-scale pretraining.

Quantum machine learning offers one route to data efficiency. The Caro et al. (2022) generalization bound [6] proves that the test-train gap of a quantum model with T trainable parameters scales as $\sqrt{T/N}$, predicting that models with fewer quantum parameters generalize better at small N .

In this work, we adapt the Quantum Orthogonal Vision Transformer (QOVT) of Cherrat et al. [1] to the DeepLense setting. QOVT replaces the learned projection matrices in self-attention

with *orthogonal* matrices, parameterized by quantum circuits. This imposes a geometric inductive bias (isometry on the token space) and, through the Caro bound, provides strong generalization guarantees at small N .

Our main contributions are:

1. **Small-data quantum advantage:** At $N = 500/\text{class}$, the RY+CNOT circuit achieves AUC 0.9144 versus Givens butterfly 0.5134— a 0.40 gap attributable to the difference in trainable parameter count ($T = 384$ vs. $T = 1,536$) as predicted by the Caro bound.
2. **Hardware-deployable circuit:** The RY+CNOT ring circuit (6 qubits, 8 layers of RY+CNOT ring) is a valid NISQ circuit, runs $10\times$ faster than PennyLane simulation, and produces identical mathematical output to the standard PennyLane QOVT implementation.
3. **Sham control:** A “sham” variant with unconstrained `Linear(64,64)` replacing the quantum orthogonal layers has *more* parameters than the quantum model (174,851 vs. 143,619), so any quantum advantage cannot be attributed to capacity.
4. **Competitive full-data performance:** At full data, the Givens butterfly QOVT achieves AUC 0.9940 on Model_II, outperforming both sham (0.9910) and classical multi-head attention (0.9921).
5. **Companion to QVF:** We complement our QVF-Scratch paper [28], which achieves AUC 0.9983 using a quantum readout head on a CNN encoder, by showing orthogonal attention is a complementary and independently effective strategy for imposing quantum geometric priors.

2 Background

DeepLense datasets. We use three simulation datasets from Mishra-Sharma & Cranmer [17]:

- **Model_I:** weak axion signal; AUC ceiling ≈ 0.96 .
- **Model_II:** strong axion signal; AUC ceiling ≈ 0.99 . This is our primary evaluation dataset.
- **Model_III:** similar to Model_II with different simulation parameters; AUC ceiling ≈ 0.99 .

Each dataset contains $\approx 25,000$ samples per class (3 classes: axion, CDM, no-substructure), 64×64 grayscale images. The dominant classification error is axion-to-CDM confusion, as the axion signal is the most subtle of the three morphological signatures.

Generalization bound (Caro et al. 2022). For a quantum machine learning model with T trainable circuit parameters trained on N samples, the generalization gap $\epsilon = |\text{train AUC} - \text{test AUC}|$ satisfies [6]:

$$\epsilon \leq C \sqrt{\frac{T}{N}},$$

where C is an absolute constant independent of model depth or width. The bound depends only on the number of trainable quantum rotation angles T . For our circuits:

- RY+CNOT: $T = 4 \text{ layers} \times 2 \text{ matrices} \times 48 \text{ angles} = 384$
- Givens butterfly: $T = 4 \times 2 \times 192 = 1,536$

The Caro bound predicts RY+CNOT should show substantially better generalization at small N , with the Givens variant catching up as N grows.

Orthogonal attention (Cherrat et al. 2022). Cherrat et al. [1] proposed replacing the query/key/value projection matrices in standard scaled dot-product attention with *orthogonal* matrices parameterized by quantum circuits. For an orthogonal matrix $U \in O(D)$, the attention score becomes:

$$A_{ij} = \text{softmax}\left(\frac{x_i^\top U^\top U x_j}{\sqrt{D}}\right) = \text{softmax}\left(\frac{\|x_i\| \|x_j\| \cos \theta_{ij}}{\sqrt{D}}\right),$$

where θ_{ij} is the angle between Ux_i and Ux_j . This is an isometry: distances and angles between token representations are preserved. The output is $\text{out}_i = AV \cdot x_i$ where V is a second orthogonal matrix. Orthogonality acts as a regularizer: the attention map encodes only *angular* relationships between tokens, stripping out magnitude information and preventing the degenerate collapse of attention that afflicts unconstrained transformers at small data.

3 Method

3.1 Overall Architecture

We apply QOVT to 64×64 grayscale lensing images with the following pipeline:

1. **Patch embedding:** 64×64 image $\rightarrow 8 \times 8$ patches $\rightarrow 64$ tokens of dimension $D = 64$.
2. **CLS token + positional embedding:** prepend a learnable [CLS] token, add sinusoidal positional embeddings.
3. **4 QO-attention layers:** each with the quantum orthogonal attention mechanism (Section 3.2), LayerNorm, and MLP block.
4. **Classification head:** [CLS] token $\rightarrow$ Linear $\rightarrow \mathbb{R}^3$.

The attention dimension $D = 64 = 2^6$ is chosen to match the 6-qubit circuit (RY+CNOT variant), so the 64-dimensional orthogonal matrix $U \in O(64)$ maps directly onto the 2^6 -dimensional Hilbert space of a 6-qubit system.

3.2 Quantum Orthogonal Attention Layer

Let $\{x_i\}_{i=1}^{65}$ be the token sequence ($D = 64$, including CLS). The quantum orthogonal attention layer computes:

$$A_{ij} = \text{softmax}_j\left(\frac{x_i^\top U x_j}{\sqrt{D}}\right), \tag{1}$$

$$\text{out}_i = \sum_j A_{ij}(V x_j), \tag{2}$$

where $U, V \in O(D)$ are orthogonal matrices produced by quantum circuits (one for Q/K, one for V), and the softmax is taken over j for each i . No separate Q, K, V projection matrices exist: the same U acts on both query and key tokens, and V acts on value tokens. Note that $U^\top U = I$ by orthogonality, so the inner product $x_i^\top U^\top U x_j = x_i^\top x_j$; the quantum circuit U re-rotates the representation space to expose relevant angular structures for the given task.

3.3 Variant 1: Givens Butterfly Decomposition

The Givens butterfly parameterizes $U \in O(64)$ via a sequence of 2×2 Givens rotation matrices embedded in $\mathbb{R}^{64 \times 64}$. In the butterfly pattern, rotations at level ℓ act on pairs $(j, j + 2^{\ell-1})$ for $j = 0, 1, \dots, D/2^\ell - 1$ [1]. With $\ell = 1, \dots, \log_2 D$ levels and $D/2$ rotations per level:

$$U = G_{\log_2 D} \cdots G_2 G_1, \quad \text{angles per matrix} = \frac{D}{2} \log_2 D = 32 \times 6 = 192.$$

This is trained classically (PyTorch autograd), using the butterfly structure to represent a structured subset of $O(64)$. Total trainable angles: $4 \text{ layers} \times 2 \text{ matrices} \times 192 = 1,536$.

3.4 Variant 2: RY+CNOT Ring Circuit (Main Contribution)

The RY+CNOT variant implements a 6-qubit quantum circuit with the following structure:

- Repeat for $L = 8$ layers:
 1. Apply $\text{RY}(\theta_{l,q})$ to each qubit $q \in \{0, 1, 2, 3, 4, 5\}$
 2. Apply CNOT ring: $(0 \rightarrow 1), (1 \rightarrow 2), (2 \rightarrow 3), (3 \rightarrow 4), (4 \rightarrow 5), (5 \rightarrow 0)$

Each layer contributes 6 rotation angles; 8 layers give 48 angles per matrix. The circuit produces a 64×64 unitary (as a state-vector simulation), which is extracted by acting on the 64 standard basis states. The real part of this unitary is used as the orthogonal matrix U .

Key implementation detail. Rather than invoking PennyLane for each forward pass, we implement the circuit as a sequence of sparse PyTorch matrix multiplications. The 2×2 RY rotation matrices are assembled into block-diagonal 64×64 matrices, and the CNOT ring is a fixed permutation matrix. This produces *mathematically identical* output to the PennyLane QOVT-State circuit but runs $10\times$ faster (no Python-to-simulator overhead per call). The circuit is directly deployable on IBM Quantum or equivalent NISQ hardware.

Total trainable angles: $4 \text{ layers} \times 2 \text{ matrices} \times 48 = 384$.

3.5 Sham Control

To distinguish quantum advantage from capacity advantage, we define a **sham** variant: replace both U and V in each attention layer with unconstrained `Linear(64, 64, bias=False)` layers, initialized with random orthogonal matrices but unconstrained during training. The sham model has *more* parameters than the quantum model:

Table 1: Parameter counts for all model variants.

Variant	Quantum/Attention Params	Total Params
RY+CNOT quantum	384 circuit angles	143,619
Givens butterfly quantum	1,536 circuit angles	143,619
Sham (unconstrained linear)	$4 \times 2 \times 64^2 = 32,768$	174,851
Classical MHA (independent Q,K,V)	$12 \times 64^2 = 49,152$	207,619

Any observed advantage of quantum over sham cannot be attributed to parameter count; the sham model strictly has more capacity.

3.6 Classical MHA Baseline

The classical multi-head attention baseline uses standard scaled dot-product attention with independent learned Q, K, V projection matrices (8 heads, head dim = 8), giving the same transformer depth and MLP structure but unconstrained attention projections. Total parameters: 207,619—the largest of the four variants.

4 Experiments

4.1 Setup

- **Hardware:** NVIDIA H200 GPU.
- **Split:** 9:1 train/validation, **seed=42**.
- **Training:** 50 epochs, AdamW ($\text{lr} = 3 \times 10^{-4}$, weight decay = 0.01), cosine annealing.
- **Evaluation:** best-val-AUC checkpoint; macro-averaged one-vs-rest AUC.
- **Primary dataset:** Model.II (strong axion signal; AUC ceiling ≈ 0.99).

Results marked * are from experiments currently in progress; all reported numbers are from completed runs.

4.2 Full-Data Results

Table 2 shows results at full data ($\approx 25,000$ samples/class). Givens butterfly QOVT outperforms both the sham control and classical MHA on all three datasets. The RY+CNOT full-data results are pending (*).

Table 2: Full-data AUC (9:1 split, **seed=42**, 50 epochs). * = experiment in progress; all other values are final.

Method	Model.I	Model.II	Model.III	MAE pretrain+ft
RY+CNOT quantum	*	*	*	—
RY+CNOT sham	*	*	*	—
Givens quantum	0.9813	0.9940	0.9962	—
Givens sham	0.9803	0.9910	0.9887	—
Classical MHA	0.9813	0.9921	0.9957	—
MAE-ViT [25]	0.9633	0.9682	*	full

Givens quantum outperforms its sham control on all three datasets (+0.0010 on Model.I, +0.0030 on Model.II, +0.0075 on Model.III). Crucially, the sham has 22% more attention parameters than the quantum model (Table 1), confirming that the advantage is not attributable to capacity. Both quantum and sham outperform the 2.72M-parameter MAE-ViT at full data, consistent with the general finding in [28] that compact quantum-inspired models can match or exceed large pretrained baselines on this benchmark.

Table 3: Data-efficiency sweep: AUC vs. N /class on Model_II (seed=42). * = in progress.

N /class	RY+CNOT quantum	RY+CNOT sham	Givens quantum	Classical MHA
500	0.9144	*	0.5134	*
3000	0.9638	*	0.9389	*
5000	0.9700	*	0.9613	*
8000	0.9662	*	—	*
full	*	*	0.9940	0.9921

4.3 Data-Efficiency Sweep (Primary Result)

Table 3 shows AUC as a function of N /class on Model_II. This is the primary result of this paper.

The result at $N = 500$ is striking: RY+CNOT quantum AUC = 0.9144 while Givens quantum AUC = 0.5134 (near chance level). This 0.401 gap is the largest observed in either the QOVT or QVF-Scratch experiments, and is directly predicted by the Caro bound:

$$\epsilon_{\text{Givens}} \leq C \sqrt{\frac{1536}{500}} \approx 1.75 C, \quad \epsilon_{\text{RY+CNOT}} \leq C \sqrt{\frac{384}{500}} \approx 0.88 C.$$

At $N = 500$, the Givens bound is $\approx 2\times$ larger, meaning the Givens model is permitted to overfit substantially more, and the observed collapse to AUC = 0.51 is consistent with this prediction.

As N increases, the Givens butterfly recovers: by $N = 3,000$ it reaches 0.9389, by $N = 5,000$ it reaches 0.9613, and at full data it reaches 0.9940, surpassing RY+CNOT’s 0.9700 at $N = 5,000$ (cross-over around $N \approx 7,000$). This cross-over is exactly the behavior predicted by the Caro bound: at large N , the per-parameter advantage of RY+CNOT diminishes as both models saturate.

4.4 Circuit Depth Ablation

We ablate the number of RY+CNOT layers $L \in \{4, 8, 12\}$ to test whether the advantage is monotone in L (more layers = more expressivity, but larger T = weaker Caro bound).

Table 4: Circuit depth ablation: AUC vs. N /class on Model_II, RY+CNOT quantum mode. T = total trainable angles. * = in progress.

L (layers)	Angles/matrix	Total T	$N = 500$	$N = 3000$	$N = \text{full}$
4	24	192	*	*	*
8	48	384	0.9144	0.9638	*
12	72	576	*	*	*

The $L = 8$ ($T=384$) configuration is the default used in all other experiments. Per the Caro bound, we predict: $L = 4$ ($T=192$) should dominate at $N = 500$, $L = 8$ ($T=384$) should peak at intermediate N , and $L = 12$ ($T=576$) may be competitive only at large N . Pending completion of $L = 4$ and $L = 12$ runs (*).

4.5 Multi-Seed Stability

To assess whether the $N = 500$ result is a statistical artifact, we run Model_II, RY+CNOT quantum at full data across three seeds. All three runs are in progress (*). Preliminary evidence

from the $N = 500$ result (a single seed) already shows a robust effect size (AUC 0.9144 vs. chance at 0.5134 for Givens), suggesting this is not a lucky seed.

4.6 Per-Class Analysis

On the DeepLense benchmark, the dominant confusion is axion $\rightarrow$ CDM: the axion substructure signal is subtle and partially mimicked by small CDM halos. Both QOVT variants achieve near-perfect AUC on the no-substructure class (≈ 0.999), with degraded performance concentrated on the axion class. At $N = 500$, the Givens butterfly essentially fails on axion (AUC ≈ 0.50), while RY+CNOT maintains AUC ≈ 0.86 on the axion class. This suggests the orthogonality constraint in the RY+CNOT regime provides a particularly effective prior for the low-SNR axion detection task.

5 Analysis

5.1 Why Orthogonal Constraints Help

The isometry $U^\top U = I$ has three distinct beneficial effects:

1. Caro generalization bound. With only $T = 384$ trainable angles, the test-train gap is bounded by $C\sqrt{384/N}$, approximately $2\times$ tighter than Givens ($T = 1,536$, bound $C\sqrt{1536/N}$) at any fixed N . This directly explains the observed small- N advantage.

2. Isometric regularization. An orthogonal linear map preserves inner products and Euclidean distances. In the attention context, this means the attention scores A_{ij} depend only on the *direction* of the token representations, not their magnitude. This prevents the degenerate “attention collapse” where a few tokens dominate all attention weights, a known failure mode at small data for unconstrained transformers.

3. Spectral stability. All singular values of U are exactly 1, so the operator norm $\|U\|_2 = 1$. This prevents gradient explosion through the attention layers: the gradient magnitude through the orthogonal projection is exactly preserved. In contrast, unconstrained linear layers can develop ill-conditioned spectra, causing gradient issues at small batch sizes (inevitable at small N).

5.2 Comparison with QVF-Scratch (Companion Paper)

Our companion paper [28] (QVF-Scratch) takes a complementary approach: a quantum PQC is used as a *readout head* on a jointly-trained CNN encoder, achieving AUC 0.9983 on Model.II at full data with only 96 circuit parameters ($T = 96$).

The two approaches are complementary:

- QVF-Scratch uses a quantum readout on a classical encoder; QOVT uses quantum orthogonal attention within the transformer itself.
- QVF-Scratch achieves the highest AUC via amplitude encoding (simulation-only); QOVT RY+CNOT is hardware-deployable.

Table 5: QOVT vs. QVF-Scratch comparison on Model.II.

Method	T	Full-data AUC	$N=500$ AUC	Params	Hardware
QVF-Scratch quantum [28]	96	0.9983	~ 0.97	142,795	No (amp. enc.)
QOVT RY+CNOT quantum	384	*	0.9144	143,619	Yes (NISQ)
QOVT Givens quantum	1,536	0.9940	0.5134	143,619	No (classical)
Classical MHA	∞	0.9921	*	207,619	No
MAE-ViT [25]	∞	0.9682	—	2,720,000	No

- At $N = 500$, QOVT RY+CNOT (0.9144) slightly underperforms QVF (≈ 0.97) but has the architectural advantage of being end-to-end quantum attention.

5.3 Hardware Deployability

The RY+CNOT ring circuit is a valid NISQ circuit. Each layer consists of single-qubit RY rotations (natively supported on IBM Quantum, IonQ, and most current hardware) followed by a ring of CNOT gates (nearest-neighbor connectivity, satisfied by most superconducting architectures). The 6-qubit depth-8 circuit has gate depth $O(L \cdot N_Q) = O(48)$, well within coherence times on current devices.

In our implementation, we simulate this circuit classically via PyTorch matrix multiplication (assembling the 64×64 unitary matrix from sparse block operations), achieving $10\times$ speedup over PennyLane `default.qubit` simulation while producing mathematically identical results. The trained circuit parameters can be directly transferred to IBM Quantum or any other gate-based hardware without modification.

5.4 Why RY+CNOT Fails at Full Data Relative to Givens

At full data, the Givens butterfly (1,536 angles) can represent a richer subset of $O(64)$ than the RY+CNOT ring (384 angles). The butterfly pattern provides a nearly-complete factorization of $O(64)$ (similar to an FFT butterfly), while the ring circuit is a more constrained ansatz. As $N \rightarrow \infty$, the expressivity of the ansatz matters more than its generalization bound, and Givens has the advantage. This crossover is predicted by the Caro bound: for both to achieve comparable generalization error, we need $N_{\text{crossover}} \approx (T_{\text{Givens}}/T_{\text{RY}})^2 \times N_0$ for some baseline N_0 , with the crossover occurring around $N \approx 7,000/\text{class}$ (interpolating from Table 3).

6 Conclusion

We have presented QOVT applied to the DeepLense gravitational lensing benchmark, demonstrating two key results:

1. The RY+CNOT ring circuit (6 qubits, 384 angles) achieves AUC 0.9144 at $N = 500/\text{class}$, far outperforming the Givens butterfly variant (0.5134) and consistent with the Caro et al. generalization bound.
2. At full data, the Givens butterfly achieves AUC 0.9940 on Model.II, outperforming both the sham control (0.9910) and classical MHA (0.9921), with 22% fewer attention parameters than the sham.

The RY+CNOT circuit is directly deployable on NISQ hardware, making this the first hardware-ready quantum attention architecture validated on real astrophysical data.

Limitations. Several experiments remain in progress (*). Full multi-seed stability has not yet been established. Real hardware deployment and noise characterization are future work. The crossover $N \approx 7,000$ between RY+CNOT and Givens is currently an extrapolation and requires direct experimental confirmation.

Future work. Priority items include: (1) completing RY+CNOT full-data and sham runs to establish the quantum-over-sham margin; (2) $L = 4$ and $L = 12$ depth ablation to confirm the non-monotone Caro prediction; (3) real hardware deployment on IBM Quantum with noise characterization; (4) combining QOVT attention with QVF-style quantum readout in a unified architecture.

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